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Secondary battery and method of manufacturing the same

Abstract

A secondary battery with a reduced degree of increase in internal pressure over time, is provided. In a preferred embodiment, a secondary battery including: an electrode body, a battery case, an electrode terminal, and a gasket sandwiched between the battery case and the electrode terminal is provided. The secondary battery is configured such that a parameter K calculated using the following equation (1): $K = W_i / (V_r G_{sub.H2})$ (1) (where W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm³) of void in the battery case, and $G_{sub.H2}$ represents an H₂ gas permeability coefficient (cm³·cm)/(cm²·s·cmHg) of the gasket at 60° C.) satisfies 0.43×10^{-8} or more to 0.59×10^{-8} or less.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
7758994	12/2009	Komori	429/185	H01M 50/30
2015/0364735	12/2014	Kohira et al.	N/A	N/A
2017/0077486	12/2016	Ishii	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2005-140196	12/2004	JP	N/A
2007-149378	12/2006	JP	N/A
2009-170159	12/2008	JP	N/A
2015-220119	12/2014	JP	N/A
6250567	12/2016	JP	N/A
2018-181544	12/2017	JP	N/A
2021-012804	12/2020	JP	N/A

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) The present application is based upon and claims the benefit of priority from Japanese patent application No. 2021-149895 filed on Sep. 15, 2021, and the entire disclosure of which is incorporated herein in its entirety by reference.

BACKGROUND OF THE DISCLOSURE

1. Technical Field

(2) The present disclosure relates to a secondary battery and a method of manufacturing the same.

2. Background

(3) Secondary batteries such as lithium-ion secondary batteries and nickel hydride batteries are widely used in various fields such as vehicles and portable terminals. This kind of secondary battery generally includes an electrode body including a positive electrode and a negative electrode, a battery case housing the electrode body, and an electrode terminal attached to the battery case and in connection with each of the electrodes of the electrode body. The secondary battery further includes a gasket sandwiched between the battery case and the electrode terminal and sealing therebetween.

(4) In the secondary battery with such a configuration, gas may be generated inside the battery case due to charging/discharging. A technology of reducing the rapid increase in the pressure inside the case due to the gas generated is described in the following two Patent Documents, for example. The disclosure of Japanese Patent No. 6250567 proposes that when the pressure inside the sealed battery is increased due to the gas generated, an exterior can is deformed from a thin portion provided in the sealing plate of the exterior can to discharge the gas generated to the outside. On the other hand, Japanese Patent Application Publication No. 2018-181544 proposes that combined use of two gaskets having different gas permeability coefficients from each other discharges the gas generated inside the battery case to the outside.

SUMMARY OF THE INVENTION

(5) The technologies disclosed in the above two Patent Documents are to release, when the pressure inside the secondary battery increases, the increased internal pressure to the outside of the battery case. In contrast, the present inventors have tried to reduce the degree of increase in the pressure inside the secondary battery over time.

(6) According to the technology disclosed herein, provided is a secondary battery including: an electrode body including a positive electrode and a negative electrode; a battery case housing the electrode body; an electrode terminal attached to the battery case and in connection with the positive electrode or the negative electrode of the electrode body; and a gasket sandwiched between the battery case and the electrode terminal and sealing between the battery case and the electrode terminal. The secondary battery **10** is configured such that a parameter K calculated using the following equation (1):

$$K = W_i / (V_r G_{H_2}) \quad (1)$$

(in the equation (1), W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm³) of void in the battery case, and G_{H_2} represents an H₂ gas permeability coefficient (cm³·cm)/(cm²·s·cmHg) of the gasket at 60° C.) satisfies 0.43×10^{-8} or more to 0.59×10^{-8} or less.

(7) With this configuration, the parameter K satisfying the range allows reduction in the degree of increase in the pressure inside the secondary battery over time. Therefore, the secondary battery can be used with a suitable battery performance for a long period of time.

(8) In a preferred aspect of the secondary battery disclosed herein, the H₂ gas permeability coefficient G_{H_2} (cm³·cm)/(cm²·s·cmHg) is 35.5×10^{-10} or more to 51.8×10^{-10} or less. With this configuration, the effect of reducing the degree of increase in the pressure inside the secondary battery over time can be exhibited more efficiently.

(9) In another preferred aspect, the secondary battery further includes a nonaqueous electrolyte including ethylene carbonate, ethyl methyl carbonate, and dimethyl carbonate. The secondary battery including a nonaqueous electrolyte with such a configuration can achieve the effects of the technology disclosed herein more efficiently.

(10) According to the technology disclosed herein, provided is a method of manufacturing a secondary battery including: an electrode body including a positive electrode and a negative electrode; a battery case housing the electrode body; an electrode terminal attached to the battery case and in connection with the positive electrode and the negative electrode of the electrode body;

and a gasket sandwiched between the battery case and the electrode terminal and sealing between the battery case and the electrode terminal. This manufacturing method includes: setting a parameter K calculated using the following equation (1):

$$K=Wi/(VrG.sub.H2) \quad (1)$$

(in the equation (1), Wi represents an initial capacity (Ah) of the secondary battery, Vr represents a volume (cm.sup.3) of void in the battery case, and G.sub.H2 Represents an H.sub.2 gas permeability coefficient (cm.sup.3.Math.cm)/(cm.sup.2.Math.s.Math.cmHg) of the gasket at 60° C.) to satisfy $0.43 \times 10^{sup.8}$ or more to $0.59 \times 10^{sup.8}$ or less to construct the secondary battery.

(11) With this configuration, the secondary battery to be manufactured is constructed by setting the parameter K to satisfy the above-described range. This allows reduction in the degree of increase in the pressure inside the secondary battery over time. Therefore, the secondary battery which can be used with a suitable battery performance for a long period of time can be provided.

(12) In the manufacturing method, the setting is preferably performed with a gasket having the H.sub.2 gas permeability coefficient G.sub.H2 (cm.sup.3.Math.cm)/(cm.sup.2.Math.s.Math.cmHg) of $35.5 \times 10^{sup.-10}$ or more to $51.8 \times 10^{sup.-10}$ or less selected as the gasket. With this setting, the effect of reducing the degree of increase in the pressure inside the secondary battery over time can be exhibited more efficiently.

(13) The manufacturing method preferably farther includes: injecting a nonaqueous electrolyte including ethylene carbonate, ethyl methyl carbonate, and dimethyl carbonate. The injecting of the nonaqueous electrolyte with such a configuration allows a secondary battery which exhibits the effect of the technology disclosed herein more suitably to be provided.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a secondary battery 10.
- (2) FIG. 2 is a cross-sectional view taken along line II-II of FIG. 1.
- (3) FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2.
- (4) FIG. 4 is a cross-sectional view taken along line IV-IV of FIG. 3.
- (5) FIG. 5 is a graph showing test results of an Example.

DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

(6) The following describes an embodiment of a secondary battery disclosed herein. The embodiment described herein is naturally not intended to limit the technology disclosed herein. Unless specifically mentioned, the technology disclosed herein is not limited to the embodiment described herein. Each drawing has been schematically illustrated and therefore may not necessarily reflect actual elements. In the drawings, the same members portions which exhibit the same action are denoted by the same reference numerals, and the duplicated descriptions may be omitted or simplified. The dimensional relation (such as length, width, or thickness) in each drawing does not reflect the actual dimensional relation. The expression “A to B” indicating a numerical range means “A or more to B or less,” and also means “above A and below B” unless otherwise specified.

(7) The “secondary battery” herein generally refers to an electricity storage device which causes a charging and discharging reaction by movement of charge carriers between a pair of electrodes (a positive electrode and a negative electrode) via an electrolyte. The “secondary battery” herein encompasses so-called secondary batteries such as a lithium-ion secondary battery, a nickel hydride battery and a nickel cadmium battery, and capacitors such as an electric double-layer capacitor. The following describes the embodiment targeting for a lithium-ion secondary battery among the secondary batteries.

(8) <<Secondary Battery 10>>

(9) FIG. 1 is a perspective view of a secondary battery 10. FIG. 2 is a cross-sectional view taken along line II-II of FIG. 1. FIG. 2 shows the state where the inside of the secondary battery 10 is exposed along one wider surface of a substantially cuboid battery case 41. The secondary battery 10 shown in FIGS. 1 and 2 is a so-called sealed battery. FIG. 3 is a cross-sectional view taken along line III-III of FIG. 2. FIG. 3 is a schematic partial cross-sectional view of the state where the inside of the secondary battery 10 is exposed along one narrower surface of a substantially cuboid battery case 41.

(10) As shown in FIGS. 1 to 3, the secondary battery 10 includes: an electrode body 20, a battery case 41, a positive electrode terminal 42, a negative electrode terminal 43, and a gasket 71.

(11) The secondary battery 10 is configured such that a parameter K calculated using the following equation (1):

$$K = W_i / (V_r G_{\text{sub.H2}}) \quad (1)$$

(in the equation (1), W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm^3) of void in the battery case, and $G_{\text{sub.H2}}$ represents an $H_{\text{sub.2}}$ gas permeability coefficient ($\text{cm}^3 \cdot \text{cm} / (\text{cm}^2 \cdot \text{s} \cdot \text{cmHg})$) of the gasket at 60°C .) satisfies 0.43×10^{-8} or more to 0.59×10^{-8} or less. The parameter K satisfying the above-described range allows reduction in the degree of increase in the pressure inside the secondary battery over time. In the secondary battery where the parameter K satisfies the above-described range, the degree of increase in the internal pressure is reduced. Thus, it is not necessary to physically open the battery case due to the increase in the internal pressure. Therefore, the secondary battery can be used with a suitable battery performance for a long period of time. In other words, durability performance of the secondary battery can be improved.

(12) The initial capacity W_i (Ah) in the equation (1) represents a battery capacity measured using the method described in the following Examples. In the measurement of the initial capacity W_i , the battery assembly immediately after assembling (i.e., before the electrode body is activated) is activated under the following conditions of the following Examples, to make the secondary battery be ready for use (activation process). This secondary battery is placed under the temperature condition at 25°C ., and charged at a constant current of $1/3 \text{ C}$ until the terminal voltage of the secondary battery reaches 4.1 V. Subsequently, the secondary battery is charged at a constant voltage until the current value reaches $1/50 \text{ C}$ to fully charge the secondary battery (SOC 100%). Then, the secondary battery is discharged at a constant current of $1/3 \text{ C}$ until the terminal voltage of the secondary battery reaches 3.0 V. At this time, the discharge capacity is measured and set as an initial capacity W_i (Ah). Note that 1 C refers to the current value which can be fully charged/discharged in one hour.

(13) The initial capacity W_i is not particularly limited as long as the parameter K satisfies the above-described range. The initial capacity W_i can be set in accordance with the desired output of the secondary battery. The initial capacity W_i thus may be set considering the balance of the volume V_r of void in the battery case and the $H_{\text{sub.2}}$ gas permeability coefficient $G_{\text{sub.H2}}$. The initial capacity W_i may be set, as appropriate, by changing the size of the electrode body, for example. Examples of the change in the size of the electrode body include changes in the total number of electrode bodies housed in the battery case, the size of the electrode, the coating weight of the electrode active material layer; and the thickness of the electrode active material layer. Alternatively, the constituent materials of the electrode body (e.g., the positive electrode active material, the negative electrode active material, and the electroconductive material) may be changed.

(14) The volume V_r (cm^3) of void in the battery case refers to a remaining space inside the battery case and can be determined by subtracting the total volume of all components housed in the battery case from the volumetric capacity of the battery case. Specifically, the volume V_r (cm^3) can be calculated using the following equation (2):

$$V_r = V_t - V_s \quad (2)$$

(in the equation (2), V_t represents the volumetric capacity of the sealed space inside the battery case, and V_s represents a total of the volume V_1 of a solid housed in the sealed space (e.g., the electrode body and attachment members of the electrode body) and the volume V_2 of the liquid (e.g., a nonaqueous electrolyte if contained). The volume V_1 of the solid is measured by the Archimedes' method (see the following Examples). The volume V_r of void in the battery case includes a pore volumetric capacity in the electrode body (i.e., the pore volumetric capacity in the electrode active material layer and the pore volumetric capacity in the separator).

(15) The volume V_r of void in the battery case is not particularly limited as long as the parameter K satisfies the above-described range. The volume V_r of void in the battery case can be set, as appropriate, according to the desired output of the secondary battery and thus may be set considering the balance of the initial capacity W_i and the $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$. The volume V_r of void in the battery case may be set, as appropriate, by changing the size of the electrode body, for example. Examples of the change in the size of the electrode body include changes in the total number of electrode bodies housed in the battery case, the size of the electrode, the coating weight of the electrode active material layer, and the thickness of the electrode active material layer. Alternatively, the constituent materials of the electrode body (e.g., the positive electrode active material, the negative electrode active material, and the electroconductive material) may be changed.

(16) Although not particularly limited thereto, the ratio (V_r/V_t) between the volume V_r of void in the battery case and the volumetric capacity V_t of the sealed space in the battery case can be set to approximately 0.1 to approximately 0.3 (e.g., 0.14 to 0.25).

(17) The $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ ($\text{cm}^3 \cdot \text{Math} \cdot \text{cm} / (\text{cm}^2 \cdot \text{Math} \cdot \text{s} \cdot \text{Math} \cdot \text{cmHg})$) is measured in accordance with HS K 7126-1 (differential pressure method). Specific procedures, measurement conditions, and measurement devices are described in detail in the following Examples.

(18) The $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ is not particularly limited as long as the parameter K satisfies the above-described range. The $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ may be set in accordance with the desired output of the secondary battery considering the balance of the initial capacity W_i and the volume V_r of void in the battery case. For example, in order to increase the output of the secondary battery, the initial capacity W_i may be increased, and the volume V_r of void in the battery case may be reduced. At this time, selection of a gasket with a larger $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ can make the parameter K in the suitable range. In terms of reducing the degree of increase in the pressure inside the secondary battery over time, the $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ may be set to the range of appropriately 20.0×10^{-10} ($\text{cm}^3 \cdot \text{Math} \cdot \text{cm} / (\text{cm}^2 \cdot \text{Math} \cdot \text{s} \cdot \text{Math} \cdot \text{cmHg})$) to approximately 75.0×10^{-10} ($\text{cm}^3 \cdot \text{Math} \cdot \text{cm} / (\text{cm}^2 \cdot \text{Math} \cdot \text{s} \cdot \text{Math} \cdot \text{cmHg})$). The $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ is set to preferably 35.5×10^{-10} ($\text{cm}^3 \cdot \text{Math} \cdot \text{cm} / (\text{cm}^2 \cdot \text{Math} \cdot \text{s} \cdot \text{Math} \cdot \text{cmHg})$) or more to 51.8×10^{-10} ($\text{cm}^3 \cdot \text{Math} \cdot \text{cm} / (\text{cm}^2 \cdot \text{Math} \cdot \text{s} \cdot \text{Math} \cdot \text{cmHg})$) or less. As will be described in detail below, the $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ may be set, as appropriate, by changing the constituent material and the like of the gasket.

(19) The size (dimensions) of the secondary battery **10** is not particularly limited. The width W (the length in the width direction X , see FIG. 1) of the secondary battery **10** may be, for example, 70 mm or more, 100 mm or more, 150 mm or more, or 200 mm or more. Alternatively, the width W may be, for example, 300 mm or less, 250 mm or less, 200 mm or less, or 150 mm or less. The length D of the secondary battery **10** in the depth direction Y (see FIG. 1) may be, for example, 5 mm or more, 7 mm or more, 10 mm or more, or 20 mm or more. Alternatively, the length D may be, for example, 30 mm or less, 25 mm or less, 20 mm or less, or 15 mm or less. The height H (the length in the height direction Z , see FIG. 1) of the secondary battery **10** may be, for example, 40 mm or more, 50 mm or more, 60 mm or more, or 70 mm or more. Alternatively, the height H may

be, for example, 80 mm or less, 75 mm or less, 70 mm or less, or 65 mm or less.

(20) <Electrode Body **20**>

(21) The electrode body **20** is housed in the battery case **41** with being covered with an insulation film (not shown) or the like. The number of electrode bodies **20** in the secondary battery **10** is not particularly limited, and may be one, or multiple (two, three, or four or more). In the present embodiment, the secondary battery **10** houses one electrode body **20**. The electrode body **20** includes a positive electrode (a positive electrode sheet **21** in FIG. 2), a negative electrode (a negative electrode sheet **22** in FIG. 2), and separator sheets **31** and **32** as a separator. The positive electrode sheet **21**, the separator sheets **31** and **32**, and the negative electrode sheet **22** are each a long strip-like member.

(22) In the positive electrode sheet **21**, positive electrode active material layers **21b** are formed on both surfaces of a positive electrode current collector foil **21a** (e.g., an aluminum foil) having a predetermined width and a predetermined thickness except for a portion **21a1** where the positive electrode current collector foil **21a** is not formed and which is set to have a certain width at one end in the width direction. Examples of the positive electrode active material included in the positive electrode active material layer **21b** include lithium transition metal oxide (such as $\text{LiNi}_{0.1/3}\text{Co}_{0.1/3}\text{Mn}_{0.1/3}\text{O}_{2}$, LiNiO_2 , LiCoO_2 , LiFeO_2 , LiMn_2O_4 , $\text{LiNi}_{0.5}\text{Mn}_{1.5}\text{O}_4$) and lithium transition metal phosphate compound (such as LiFePO_4). The positive electrode active material layer **21b** may contain a component such as an electroconductive material and a binder, besides the active material. The electroconductive material used can be, for example, carbon black such as acetylene black (AB) and other carbon materials (such as graphite). The binder used may be, for example, polyvinylidene fluoride (Pvdf).

(23) In the negative electrode sheet **22**, negative electrode active material layers **22b** containing a negative electrode active material are formed on both surfaces of a negative electrode current collector foil **22a** (here, a copper foil) having a predetermined width and a predetermined thickness except for a portion **22a1** where the negative electrode current collector foil **22a** is not formed and which is set to have a certain width at one end in the width direction. Examples of the negative electrode active material contained in the negative electrode active material layer **22b** include carbon materials such as graphite, hard carbon, and soft carbon. The negative electrode active material layer **22b** may further contain a component such as a binder and a thickener, besides the active material. Examples of the binder used include styrene-butadiene rubber (SBR). Examples of the thickener used include carboxymethyl cellulose (CMC).

(24) The separator sheets **31** and **32** used may each be a porous resin sheets through which an electrolyte with a desired heat resistance can pass. Examples of the separator sheets **31** and **32** include a porous sheet (films) made of resin such as polyethylene (PE), polypropylene (PP), polyester, cellulose, and polyamide. Such a porous sheet may have a monolayer structure, or a lamination structure of two or more layers (e.g., a three-layer structure where PP layers are stacked on both surfaces of a PE layer). The surface of the separator sheets **31** and **32** may be provided with a heat-resistant layer (HRL) made of ceramic particles.

(25) The negative electrode active material layer **22b** is formed to have a width greater than the width of the positive electrode active material layer **21b**, for example. The widths of the separator sheets **31** and **32** are greater than that of the negative electrode active material layer **22b**. The portion **21a1** where the positive electrode current collector foil **21a** is not formed and the portion **22a1** where the negative electrode current collector foil **22a** is not formed are disposed to face each other in the width direction. The positive electrode sheet **21**, the separator sheet **31**, a negative electrode sheet **22**, and a separator sheet **32** are aligned in the length direction, and are wound up in turn on top of each other. The negative electrode active material layer **22b** covers the positive electrode active material layer **21b** with the separator sheets **31** and **32** interposed therebetween. The negative electrode active material layer **22b** is covered with the separator sheets **31** and **32**.

The portion **21a1** where the positive electrode current collector foil **21a** is not formed protrudes from one side of the separator sheets **31** and **32** in the width direction. The portion **22a1** where the negative electrode current collector foil **22a** is not formed protrudes from the separator sheets **31** and **32** the other side in the width direction.

(26) As shown in FIG. 2, the above-mentioned electrode body **20** is a flat along one plane including the winding axis so as to be housed in the case body **41a** of the battery case **41**. The portion **21a1** where the positive electrode current collector foil **21a** is not formed is disposed on one side, and the portion **22a1** where the negative electrode current collector foil **22a** is not formed is disposed on the other side, of the electrode body **20** along the winding axis.

(27) <Battery Case **41**>

(28) As shown in FIG. 2, the battery case **41** houses the electrode body **20**. As shown in FIGS. 1 and 2, the battery case **41** includes a case body **41a** having an opening in one side surface and a substantially cuboid square shape, and a lid **41b** attached to the opening. In this embodiment, the case body **41a** and the lid **41b** are formed of aluminum or an aluminum alloy mainly containing aluminum in order to reduce weight and ensure the required rigidity.

(29) <Case Body **41a**>

(30) The case body **41a** has an opening in one side surface and a substantially cuboid square shape. The case body **41a** has a substantially rectangular bottom surface **61**, a pair of wider surfaces **62** and **63** (see FIG. 3), and a pair of narrower surfaces **64** and **65**. The pair of wider surfaces **62** and **63** are standing from the longer side of the bottom surface **61**. The pair of narrower surfaces **64** and **65** are standing from the shorter side of the bottom surface **61**. In one side surface of the case body **41a**, an opening **41a1** surrounded by the pair of wider surfaces **62** and **63** and the pair of narrower surfaces **64** and **65** is formed.

(31) <Lid **41b**>

(32) The lid **41b** is attached to the opening **41a1** in the case body **41a** surrounded by longer sides of the pair of wider surfaces **62** and **63** (see FIG. 3) and shorter sides of the pair of narrower surfaces **64** and **65**. The outer edge of the lid **41b** is bonded to the edge of the opening **41a1** of the case body **41a**. Such bonding may be performed, for example, by continuous welding without gaps. Such welding can be achieved, for example, by laser welding. Although detailed illustrations are omitted, the lid **41b** can be provided with a liquid injection hole and a gas discharge valve. The liquid injection hole is for injecting a nonaqueous electrolyte after bonding the lid **41b** to the case body **41a**. The liquid injection hole is sealed with a sealing member. The gas discharge valve is a thin portion configured to be broken and to discharge gas inside the battery case **41** to the outside when the pressure inside the battery case **41** exceeds a predetermined value.

(33) The positive electrode terminal **42** and the negative electrode terminal **43** are attached to the battery case **41** (the lid **41b** in this embodiment), and are in connection with the respective electrodes of the electrode body **20**. The positive electrode terminal **42** includes an inner terminal **42a** and an outer terminal **42b**. The negative electrode terminal **43** includes an inner terminal **43a** and an outer terminal **43b**. The inner terminals **42a** and **43a** are attached to the inside of the lid **41b** via an insulator **72**. The outer terminals **42b** and **43b** are attached to the outside of the lid **41b** via a gasket **71**. The inner terminals **42a** and **43a** extend into the case body **41a**. The inner terminal **42a** of the positive electrode is in connection with the portion where the positive electrode current collector foil **21a** is not formed. The inner terminal **43a** of the negative electrode is in connection with a portion **22a1** where the negative electrode current collector foil **22a** is not formed.

(34) As shown in FIG. 2, the portion **21a1** where the positive electrode current collector foil **21a** of the electrode body **20** is not formed and the portion **22a1** where the negative electrode current collector foil **22a** of the electrode body **20** is not formed are attached respectively to the inner terminals **42a** and **43a** attached to both sides of the lid **41b** in the longitudinal direction. The electrode body **20** is housed in the battery case **41** with being attached to the inner terminals **42a** and **43a** attached to the lid **41b**.

(35) Although not shown in the drawings, the secondary battery **10** further includes a nonaqueous electrolyte. In other words, the secondary battery **10** may be a nonaqueous electrolyte secondary battery including a nonaqueous electrolyte. The battery case **41** may house a nonaqueous electrolyte together with the electrode body **20**. Gas (e.g., carbon dioxide) may generate inside the battery case when the nonaqueous electrolyte secondary battery is charged/discharged, overcharged, or stored in a high-temperature environment, thereby increasing the pressure inside the battery. In other words, the effects of the technology disclosed herein may be applied particularly preferably to the aspect where the secondary battery **10** includes a nonaqueous electrolyte. Although not particularly limited thereto, the nonaqueous electrolyte used may be obtained by dissolving a supporting electrolyte in a nonaqueous solvent. Examples of the nonaqueous solvent include carbonate-based solvents such as ethylene carbonate, dimethyl carbonate, and ethyl methyl carbonate. Such nonaqueous electrolytes may be used alone or in combination of two or more of them. Examples of the supporting electrolyte include fluorine-containing lithium salts such as LiPF₆. The nonaqueous electrolyte may contain various additives such as a film forming agent.

(36) FIG. **4** is a cross-sectional view taken along line IV-IV of FIG. **3**. FIG. **4** shows the cross section of a portion where the negative electrode terminal **43** is attached to the lid **41b**. As shown in FIG. **4**, the lid **41b** has an attachment hole **41b1** attached to the outer terminal **43b** of the negative electrode. The attachment hole **41b1** penetrates the lid **41b** at a predetermined position of the lid **41b**. The inner terminal **43a** and the outer terminal **43b** of the negative electrode are attached to the attachment hole **41b1** of the lid **41b** with a gasket **71** and an insulator **72** interposed therebetween. The outside of the attachment hole **41b1** is provided with a step **41b2** for attaching the gasket **71** to the periphery of the attachment hole **41b1**. The step **41b2** is provided with a seat surface **41b3** on which the gasket **71** is disposed. The seat surface **41b3** is provided with a projection **41b4** for positioning the gasket **71**.

(37) The outer terminal **43b** of the negative electrode includes a head **43s** and a shaft **43t**, as shown in FIG. **4**. In the embodiment shown in FIG. **4**, the head **43s** and the shaft **43t** are made of different members from each other. The head **43s** is a portion disposed outside the lid **41b**. The head **43s** is a substantially flat plate-like portion larger than the attachment hole **41b1** and has a recess **43s1**. The shaft **43t** is a portion attached to the attachment hole **41b1** via the gasket **71**. The shaft **43t** protrudes downward from a substantially center portion of the head **43s**. The shaft **43t** has a shaft body **43t1** and a flange portion **43t2** extending from one end of the shaft body **43t1** in the outer diameter direction. The flange portion **43t2** has a substantially circular shape, and fitted in the recess **43s1** of the head **43s**. The shaft body **43t1** is further provided with a crimping portion **43t3** to be crimped in the inner terminal **43a** on the side opposite to the side on which the flange portion **43t2** is provided. As shown in FIG. **4**, the crimping portion **43t3** is crimped in the lid **41b** by the inner terminal **43a** of the negative electrode. The crimping portion **43t3** extends from the shaft body **43t1**, inserted into the lid **41b**, and then bent and crimped by the inner terminal **43a** of the negative electrode.

(38) Although not particularly limited thereto, the head **43s** and the shaft **43t** may be made of different metals from each other in terms of connectivity with the busbar. As a result, when the busbar is made of aluminum, the metal configuring the head **43s** may be aluminum or an aluminum alloy, and the metal configuring the shaft **43t** may be copper or a copper alloy.

(39) <Gasket **71**>

(40) The gasket **71** is a member sandwiched between the battery case **41** and the electrode terminal (here, the negative electrode terminal **43**) and sealing between the battery case **41** and the electrode terminal. Specifically, as shown in FIG. **4**, the gasket **71** is attached to the attachment hole **41b1** and the seat surface **41b3** of the lid **41b**. In this embodiment, the gasket **71** includes a seat portion **71a**, a boss **71b**, and a side wall **71c**. The seat portion **71a** is a portion attached to the seat surface **41b3** provided on the outer surface around the attachment hole **41b1** of the lid **41b**. The seat portion

71a has a substantially flat surface to match the seat surface **41b3**. The seat portion **71a** has a recess corresponding to the projection **41b4** of the seat surface **41b3**. The boss **71b** protrudes from the bottom surface of the seat portion **71a**. The boss **71b** has an outer shape along the inner surface of the attachment hole **41b1** so as to be attached to the attachment hole **41b1** of the lid **41b**. The inner surface of the boss **71b** forms an attachment hole to which the shaft **43t** of the outer terminal **43b** is attached. The side wall **71c** is standing from the periphery of the seat portion **71a**. The head **43s** of the outer terminal **43b** is attached to the site surrounded by the side wall **71c** of the gasket **71**.

(41) The gasket **71** is disposed between the lid **41b** and the outer terminal **43b**, and ensures insulation between the lid **41b** and the outer terminal **43b**. The gasket **71** ensures the airtightness of the attachment hole **41b1** in the lid **41b**. In light of this, resin materials with excellent chemical resistance and weather resistance may be suitably used. Examples of the resin material include fluorine resins such as perfluoroalkoxy alkane (PFA) (e.g., a copolymer of tetrafluoroethylene and perfluoroalkoxyethylene (e.g., perfluoroalkylvinylether) (e.g., tetrafluoroethylene-perfluoroalkylvinylether copolymer, polytetrafluoroethylene (PTFE), a perfluoroethylene propene copolymer (FEP), ethylene-tetrafluoroethylene copolymer (ETFE), and polychlorotrifluoroethylene (PCTFE); and polyolefin resins such as polypropylene (PP) and polyethylene (PE). The kind of the resin material configuring the gasket **71** may be selected such that, for example, the H.sub.2 gas permeability coefficient G.sub.H2 satisfies the above-described range.

(42) The method of setting the gas permeability coefficient G.sub.H2 to a desired numerical value may be changing of the resin material constituting the gasket **71** or changing, as appropriate, the settings of the specific gravity and the mixing ratio of additives such as a filler material.

(43) <Insulator **72**>

(44) The insulator **72** is a member that is attached to the inside of the lid **41b** around the attachment hole **41b1** of the lid **41b**. The insulator **72** includes a base **72a**, a hole **72b**, and a side wall **72c**. The base **72a** is a portion disposed along the inner surface of the lid **41b**. In the present embodiment, the base **72a** is a substantially flat plate-like portion. The base **72a** is disposed along the inner surface of the lid **41b**, and is large enough not to protrude from the lid **41b** so as to be housed in the case body **41a**. The hole **72b** is provided to correspond to the inner surface of the boss **71b** of the gasket **71**. In the present embodiment, the hole **72b** is provided in the substantially center portion of the base **72a**. On the side surface facing the inner surface of the lid **41b**, a recessed step **72b1** is provided around the hole **72b**. The tip of the boss **71b** of the gasket **71** attached to the attachment hole **41b1** is fitted in the step **72b1** so that the step **72b1** does not interfere. The side wall **72c** is standing from the outer edge of the base **72a**. A basal portion **43a1** provided at one end of the inner terminal **43a** of the negative electrode is fitted to the base **72a**. The insulator **72** suitably has an appropriate chemical resistance because of being disposed inside the battery case **41**. The constituent material of the insulator **72** used may be, for example, poly phenylene sulfide resin (PPS). The material for the insulator **72** is not limited to PPS.

(45) The inner terminal **43a** of the negative electrode includes a basal portion **43a1** and a connection piece **43a2** (see FIGS. 2 and 3). The basal portion **43a1** is a portion attached to the base **72a** of the insulator **72**. In this embodiment, the basal portion **43a1** has a shape corresponding to the inner side of the side wall **72c** around the base **72a** of the insulator **72**. The connection piece **43a2** extends from one end of the basal portion **43a1** into the case body **41a** to be connected to the portion **22a1** of the electrode body **20** where the negative electrode is not formed (see FIGS. 2 and 3).

(46) In this embodiment, the gasket **71** is attached in the outside of the lid **41b** while the boss **71b** is attached to the attachment hole **41b1**. The outer terminal **43b** is attached to the gasket **71**. In this case, the shaft **43t** of the outer terminal **43b** is inserted into the boss **71b** of the gasket **71**, and the head **43s** of the outer terminal **43b** is disposed on the seat portion **71a** of the gasket **71**. The insulator **72** and the negative electrode terminal **43** are attached to the inside of the lid **41b**. As shown in FIG. 4, the crimping portion **43t3** of the outer terminal **43b** is bent and crimped to the

basal portion **43a1** of the negative electrode terminal **43**. The crimping portion **43t3** of the outer terminal **43b** and the basal portion **43a1** of the negative electrode terminal **43** may be partially bonded to each other by welding or metal bonding to improve electroconductivity.

(47) As mentioned above, the attachment of the electrode terminal to the lid **41b** is described using the negative electrode terminal **43** as an example. For the positive electrode, as an outer terminal **42b**, an outer terminal having a head and a shaft made of aluminum or an aluminum alloy may be used. The other structure is basically the same as that on the negative electrode side. Thus, the detailed description is omitted here.

(48) When gaskets of different compositions are used for the positive and negative electrodes sides, the arithmetic means of the H.sub.2 gas permeability coefficient G.sub.H2p of the gasket on the positive electrode side at 60° C. and the H.sub.2 gas permeability coefficient G.sub.H2n of the gasket on the negative electrode side at 60° C. can be the H.sub.2 gas permeability coefficient G.sub.H2 in the equation (1).

(49) That is, in the above case, the H.sub.2 gas permeability coefficient G.sub.H2 can be calculated using the following equation (3):

$$\text{H.sub.2 gas permeability coefficient G.sub.H2} = (\text{H.sub.2 gas permeability coefficient G.sub.H2p} + \text{H.sub.2 gas permeability coefficient G.sub.H2n}) / 2 \quad (3)$$

<<Method of Manufacturing Secondary Battery **10**>>

(50) The method of manufacturing the secondary, battery **10** according to an embodiment will be described below with reference to FIGS. **1** to **4**, as appropriate. In the method of manufacturing the secondary battery **10**, first, an electrode body **20**, a battery case **41**, a positive electrode terminal **42**, a negative electrode terminal **43**, and a gasket **71** are provided. Then, the secondary battery **10** is constructed such that a parameter K calculated using the following equation (1):

$$K = W_i / (V_r G_{\text{sub.H2}}) \quad (1)$$

(in the equation (1), W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm³) of void in the battery case, and $G_{\text{sub.H2}}$ represents an H.sub.2 gas permeability coefficient (cm³·cm)/(cm²·s·cmHg) of the gasket at 60° C.) is set to satisfy 0.43×10^{-8} or more to 0.59×10^{-8} or less. The manufacturing method described above allows a secondary battery where the degree of increase in the internal pressure over time is reduced and the durability performance is improved to be provided.

(51) The parameter K can be set by setting, as appropriate, the initial capacity W_i of the secondary battery, the volume V_r of void in the battery case, and the H.sub.2 gas permeability coefficient $G_{\text{sub.H2}}$ of the gasket at 60° C. as described above. This allows reduction in the degree of increase in the pressure inside the secondary battery over time and improvement of the durability performance. In addition to such effects, effects such as improvement in the output of the secondary battery and the like can be realized by setting the elements.

(52) The method of manufacturing the secondary battery **10** includes, as an example, an electrode body provision step, an electrode body attachment step, an electrode body housing step, a liquid injecting step, and a sealing step. In the electrode body provision step, an electrode body **20** is provided. In this step, the electrode body **20** designed such that the parameter K is in the suitable range is provided. Specifically, an electrode body set to have an initial capacity W_i which can realize the desired output by changing, as appropriate, the size of the electrode body and the constituent materials of the electrode body is provided. Except for this point, the method of producing the electrode body **20** itself can be any well-known method.

(53) In the electrode body attachment step, an electrode terminal is attached to the electrode body **20**. Specifically, for example, first, a lid **41b** of a battery case **41** with a positive electrode terminal **42** and a negative electrode terminal **43** attached is provided. Subsequently, the electrode body **20** is attached to an integrated body of such an lid **41b** and the electrode terminal. The portion of the electrode body **20** to which the electrode terminal is attached is as mentioned above. Thus, the description thereof is omitted herein. The gasket having an H.sub.2 gas permeability coefficient

G.sub.H2 in the specific range, selected in the setting above, may be used.

(54) In the electrode body housing step, the electrode body **20** connected to the lid **41b** via the electrode terminal is housed in the case body **41a**, and the lid **41b** and the case body **41a** are bonded to each other. Subsequently, the nonaqueous electrolyte is injected from the liquid injection hole provided in the lid **41b** (the liquid injecting step). Then, the liquid injection hole is sealed with a sealing member (the sealing step), and an activation process is conducted under predetermined conditions. Thus, a secondary battery **10** which is ready to use can be produced.

(55) The secondary battery **10** can be used in various applications. Suitable applications include power sources for driving, to be mounted on vehicles such as battery electric vehicles (BEVs), hybrid electric vehicles (HEVs), and plug in hybrid electric vehicles (PHEVs). Further, the secondary battery **10** can be used as a storage battery such as a small power storage device. Typically, the multiple secondary batteries **10** used may be connected in series and/or parallel to be in an assembled battery.

(56) In the embodiment, a secondary battery **10** including a flat wound electrode body as an example of an electrode body **20** has been described above. However, the secondary battery disclosed herein can be configured as a secondary battery including a laminated electrode assembly (i.e., an electrode assembly where multiple positive electrodes and multiple negative electrodes are stacked alternately). The secondary battery may be configured as a cylindrical secondary battery, a laminated secondary battery, or the like.

EXAMPLES

(57) Some examples regarding the present disclosure will be described below. However, it is not intended that the present disclosure is limited to such examples.

(58) <Provision of Gasket>

(59) Two kinds of gaskets including a gasket R and a gasket S were provided to produce a secondary battery for evaluation.

(60) —Gasket R—

(61) Trade name and distributor: AP230 (Daikin Industries, Ltd.) Material: PFA (a copolymer of tetrafluoroethylene and tetrafluoroethylene-perfluoroalkylvinylether) Specific gravity: 2.14

—Gasket S— Trade name and distributor: NP-20 (Daikin Industries, Ltd.) Material: FEP (a copolymer of tetrafluoroethylene and hexafluoropropylene) Specific gravity: 2.15

<Measurement of H.sub.2 Gas Permeability Coefficient G.sub.H2>

(62) For measurement of the H.sub.2 gas permeability coefficient G.sub.H2, a disk-shaped test pieces (n=3) with a thickness of 0.2 mm and a diameter of 70 mm to 150 mm were provided by using the gasket R and the gasket S. By using these test pieces, the H.sub.2 gas permeability coefficient G.sub.H2 (cm.sup.3.Math.cm)/(cm.sup.2.Math.s.Math.cmHg) of each test piece in the thickness direction was measured in accordance with HS K7126-1 “Plastics—Film and sheeting—Determination of gas-transmission rate Part 1: Differential-pressure methods.” As a measurement device, a gas-transmission tester “MT-C3” manufactured by Toyo Seiki Seisaku-sho, Ltd. was used. The measurement conditions were as follows: a measurement temperature: 60° C., the transmission cross section of the measurement cell: 38.46 cm.sup.2, the differential pressure: 800 mmHgG to 850 mmHgG on the primary side, and 0 mmHgG on the detection side. The measurement was performed in a room at 25° C. The measurement was conducted on three test pieces, and the arithmetic means of the obtained measurement values was used as the H.sub.2 gas permeability coefficient G.sub.H2 of each of the gasket R and the gasket S. The H.sub.2 gas permeability coefficient G.sub.H2 of the gasket R was 35.5×10.sup.-10

(cm.sup.3.Math.cm)/(cm.sup.2.Math.s.Math.cmHg), and the H.sub.2 gas permeability coefficient G.sub.H2 of the gasket S was 51.8×10.sup.-10

(cm.sup.3.Math.cm)/(cm.sup.2.Math.s.Math.cmHg).

Example 1

(63) <Production of Secondary Battery for Evaluation>

(64) LiNi.sub.1/3Co.sub.1/3Mn.sub.1/3O.sub.2 (LNCM) as a positive electrode active material powder, acetylene black (AB) as an electroconductive material, and polyvinylidene fluoride (PVdF) as a binder were mixed at a mass ratio of LNCM:AB:PVdF=90:8:2 in N-methylpyrrolidone (NMP). Thus, a slurry for forming a positive electrode active material layer was prepared. The slurry was applied to both sides of a long aluminum foil (the length in the longitudinal direction: 360 cm, the length in the lateral direction: 105 cm) in strip form (the coating amount: 11.0 mg/cm.sup.2), and then dried and pressed to produce a positive electrode sheet.

(65) In ion-exchange water, natural graphite (C) as a negative electrode active material, styrene-butadiene rubber (SBR) as a binder, and carboxymethyl cellulose (CMC) as a thickener were mixed at a mass ratio C:CBR:CMC=98:1:1. Thus, a slurry for forming a negative electrode active material layer was prepared. The slurry was applied to both sides of a long copper foil (the length in the longitudinal direction: 370 cm, the length in the lateral direction: 107 cm) in strip form (the coating amount: 7.0 mg/cm.sup.2), and then dried and pressed to produce a negative electrode sheet.

(66) As a separator, two porous polyolefin sheets having a three-layer structure of PP/PE/PE were provided. The positive and negative electrode sheets produced as described above and two separator sheets provided above were stacked, then wound, and thereafter pressed from the side to abduct them to produce a flat wound electrode body. Subsequently, positive and negative electrode terminals and a lid of the battery case are attached to this electrode body, which were then housed in a case body. The lid and the case body were sealed by laser welding. Here, the gasket R was used for the attachment between the lid and the electrode terminals. As the battery case, a square battery case A having a liquid injection hole for the nonaqueous electrolyte was used. As dimensions of the battery case A, the width W was 120 mm, the length D was 12.6 mm, and the height H was 65 mm (see FIG. 1).

(67) Subsequently, a volume V2 (cm) of the nonaqueous electrolyte was injected through the liquid injection hole of the battery case, and the liquid injection hole was hermetically sealed. As the nonaqueous electrolyte, one obtained by dissolving LiPF.sub.6 with a concentration of 1.0 mol/L as a supporting electrolyte in a mixed solvent containing ethylene carbonate (EC), dimethyl carbonate (DMC), and ethyl methyl carbonate (EMC) at a volume ratio EC:EMC:DMC=3:4:3 was provided.

(68) The secondary battery assembly after the injection of the nonaqueous electrolyte was stabilized in an environment at 25° C. and subjected to activation process. As the activation process, the secondary battery assembly was charged at a constant current up to 4.1 V at 25° C., and then subjected to aging process for 20 hours in an environment at 60° C. In this manner, a secondary battery for evaluation according to Example 1 was prepared.

(69) <Measurement of Initial Capacity Wi>

(70) After the activation process, the initial capacity Wi of the secondary battery for evaluation was measured. Specifically, the secondary battery for evaluation after the activation process was placed under the temperature condition at 25° C., and charged at a constant current of 1/3 C until the terminal voltage of the secondary battery reached 4.1 V. Subsequently, the secondary battery for evaluation was charged at a constant voltage until the current value reached 1/50 C to fully charge the secondary battery (SOC 100%). Then, the secondary battery for evaluation was discharged at a constant current of 1/3 C until the terminal voltage of the secondary battery reached 3.0 V. At this time, the discharge capacity was measured and set as an initial capacity Wi (Ah). The results are shown in the corresponding column in Table 1.

(71) <Measurement of Volume Vr of Void in Battery Case>

(72) The volume Vr of void in the battery case was measured by the Archimedes' method by using the secondary battery assembly after sealing between the lid and the case body and before injection of the nonaqueous electrolyte. Specifically, the nonaqueous electrolyte was injected to the secondary battery assembly to fill the battery case, and the volume of the nonaqueous electrolyte injected was recorded. A value obtained by subtracting this volume from the volumetric capacity Vt of the battery case was set as a volume V1 of the solid housed in the sealed space of the battery

case. Then, the volume V_r of void in the battery case was calculated using the following equation (4):

$$V_r = V_t - (V_1 + V_2) \quad (4)$$

V_2 represents, as described above, the volume of the nonaqueous electrolyte. The results are shown in the corresponding column in Table 1.

<Calculation of Parameter K>

(73) A parameter K was calculated using the equation (1) based on the initial capacity W_i , the volume V_r of void in the battery case, and the $H_{sub.2}$ gas permeability coefficient $G_{sub.H2}$ obtained as described above. The results are shown in the corresponding column in Table 1.

Examples 2, 3, and 8 to 10

(74) Secondary batteries for test according to the Examples were prepared in the same manner as in Example 1 except that the configuration of the electrode body (the coating weights of the slurries for forming electrode active material layers, the thicknesses of the electrode active material layers, the lengths of the aluminum foil and the copper foil) was changed, as appropriate, and the initial capacity W_i and the volume V_r of void in the battery case were adjusted (calculated by using the same method as in Example 1, see the corresponding column in Table 1), and the parameter K was set to a value described in the corresponding column in Table 1.

Examples 4, 5, 7, 12, and 13

(75) Gasket S was used instead of gasket R. Secondary batteries for test according to the Examples were prepared in the same manner as in Example 1 except that the configuration of the electrode body (the coating weights of the slurries for forming electrode active material layers, the thicknesses of the electrode active material layers, the lengths of the aluminum foil and the copper foil) was changed, as appropriate, and the initial capacity W_i and the volume V_r of void in the battery case were adjusted (calculated by using the same method as in Example 1, see the corresponding column in Table 1), and the parameter K was set to a value described in the corresponding column in Table 1.

Examples 6 and 11

(76) A battery case B was used instead of the battery case A. Secondary batteries for test according to the Examples were prepared in the same manner as in Example 1 except that the configuration of the electrode body (the coating weights of the slurries for forming electrode active material layers, the thicknesses of the electrode active material layers, the lengths of the aluminum foil and the copper foil) was changed, as appropriate, and the initial capacity W_i and the volume V_r of void in the battery case were adjusted (calculated by using the same method as in Example 1, see the corresponding column in Table 1), and the parameter K was set to a value described in the corresponding column in Table 1. As dimensions of the battery case B, the width W was 137 mm, the length D was 13.3 mm, and the height H was 63.2 mm (see FIG. 1).

(77) <Measurement of Internal Pressure Increase Slope S>

(78) The SOC of each of the secondary batteries for test according to the Examples was adjusted to 91%. The projection of a differential pressure gauge (PRESSURE GAUGE manufacturer: Limited company Nagoya Seiki Seisakusyo) was inserted into each battery case to measure the internal pressure P_1 (MPa) before storage. Then, the secondary battery for test was placed in a thermostatic chamber set at 60° C. and stored for 120 days. After 120 days, the internal pressure P_2 (MPa) after storage was measured using the differential pressure gauge. Thereafter, the internal pressure increase slope S was calculated using the following equation (5):

$$S = (P_2 - P_1) / \sqrt{\text{day}} \quad (5)$$

The results are shown in the corresponding column in Table 1. In this example, when the internal pressure increase slope S calculated as described above was 0.25×10^{-2} (MPa/ $\sqrt{\text{day}}$) or less, the degree of increase in the pressure inside the secondary battery for evaluation over time was evaluated as being sufficiently low.

<Storage Test>

(79) The SOC of each of the secondary batteries for test according to the Examples was adjusted to 80%, the secondary battery was then subjected to a storage test of storing it for a predetermined period of time under the temperature condition at 75° C., and thereafter the capacity Wc of the secondary battery for evaluation (the capacity after storage) was measured. Then, a value determined using the following equation (6):

$$\text{Capacity retention rate (\%)} = (Wc/Wi) \times 100 \quad (6)$$

was set as a capacity retention rate (%) of each Example. The results are shown in the corresponding column in Table 1. In this example, when the capacity retention rate calculated as described above was 98.0% or more, the secondary battery for evaluation was evaluated as having high durability performance.

(80) TABLE-US-00001 TABLE 1 H.sub.2 Gas Permeability Internal Pressure Initial Volume Vr of Coefficient G.sub.H2 Increase Slope S Capacity Parameter Capacity Void in Case ($\times 10^{\text{sup.}-10}$) (cm³ .Math. cm)/ Battery ($\times 10^{\text{sup.}-2}$) Retention Ex. K ($\times 10^{\text{sup.}8}$) Wi (Ah) (cm^{sup.3}) (cm^{sup.2} .Math. s .Math. cmHg) Case (Mpa/√ day) Rate (%)

1	0.59	4.25	20.1	35.5	A	0.21	98.1	2	0.56	4.22	21.2	0.18	98.4	3	0.48	4.20	24.4	0.15	98.0	4	0.58	4.15	13.8	51.8	0.22	98.6	5	0.55	4.13	14.6	0.16	98.4	6	0.57	4.32	21.5	35.5	B	0.20	98.1	7	0.43	4.39	19.8	51.8	A	0.15	98.1	8	0.88	4.23	13.5	35.5	0.62	98.4	9	0.80	4.16	14.7	0.61	98.2	10	0.67	4.05	17.0	0.52	98.3	11	0.62	4.42	20.2	B	0.32	97.9	12	0.38	4.26	21.5	51.8	A	0.12	97.5	13	0.32	4.20	25.2	0.10	96.9
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(81) FIG. 5 is a graph showing test results of an Example. In FIG. 5, the vertical axis indicates the internal pressure increase slope S ($\times 10^{\text{sup.}-2}$) (MPa/√day) (the left side in FIG. 5) and the capacity retention rate (%) (the right side in FIG. 5), and the horizontal axis indicates a parameter K ($\times 10^{\text{sup.}8}$). In FIG. 5, the marker “square-solid.” indicates the internal pressure increase slope S of Examples 1 to 7, the marker “o” indicates the internal pressure increase slope S of Examples 8-13, and the marker “x” indicates the capacity retention rate of each Example. In an area surrounded by dotted lines A and B in FIG. 5 shows a range of the parameter K when each secondary battery was evaluated as having “sufficiently small degree of increase in internal pressure over time” and “high durability performance.”

(82) As can be seen from the results shown in Table 1 and FIG. 5, in the secondary batteries (Examples 1 to 7) including an electrode body which includes a positive electrode and a negative electrode, a battery case housing the electrode body, electrode terminals attached to the battery case and in connection with electrodes of the electrode body, and gaskets sandwiched between the battery case and each electrode terminal and sealing between the battery case and each electrode terminal and configured such that a parameter K calculated using the equation (1):

$K = Wi / (Vr \cdot G_{\text{sub.H2}})$ satisfies $0.43 \times 10^{\text{sup.}8}$ or more to $0.59 \times 10^{\text{sup.}8}$ or less, the internal pressure increase slope S was $0.25 \times 10^{\text{sup.}-2}$ (MPa/√day) or less. In other words, it was demonstrated that the degree of increase in the pressure in such a secondary battery over time was sufficiently reduced. Further, in the secondary batteries for evaluation according to Examples 1 to 7, the capacity retention rate after the storage test was 98.0% or more. In other words, it was demonstrated that improvement in the durability performance was achieved in such a secondary battery. As can be seen from examples using the secondary batteries for evaluation according to Examples 8 to 13, the secondary battery where the parameter K does not satisfy the above-described range did not achieve both the reduction in the degree of increase in the internal pressure over time and the improvement in the durability performance.

(83) As can be seen from the comparison between Examples 1 to 5 and 7 using the battery case A and Example 6 using the battery case B, the secondary battery where the parameter K satisfies the specific range reduced the degree of increase in the internal pressure over time and improved durability performance regardless of the size of the battery case. The internal pressure increase slope S is determined by the difference between gas generated inside the battery case and gas discharged to the outside of the battery case and the volume of gas retained inside the battery case. In other words, the amount (amount of substance) of gas generated inside the battery case increases

as the capacity of the secondary battery increases. Further, the amount (amount of substance) of gas discharged to the outside of the battery case increases as the transmission coefficient of the gasket increases. The volume of gas retained inside the battery case may change depending on the size of the battery case. The state equation of gas expressed using the amount of substance, the volume, and the pressure (internal pressure) is constant. In order to maintain the volumetric energy density of the secondary battery, the secondary battery is generally developed so that the volume of void in the battery case (i.e., dead space) in the battery case is reduced. Therefore, it is considered that even if the size of the battery case changes, the above-described effect can be obtained when the parameter K satisfies the specific range.

(84) As can be seen from comparison between Examples 5 and 6 and Examples 8 and 9, even if the volume V_r of void in the case is smaller than that in other examples of the present Examples, the parameter K can be in the suitable range by changing the H.sub.2 gas permeability coefficient G_{H_2} of the gasket used. In other words, it was demonstrated that effective parameter K can be achieved even if the occupied volume of the electrode body in the battery case is increased for higher output (i.e., the volume of void in the base is required to be reduced).

(85) Although specific examples of the technology disclosed herein have been described in detail above, they are mere examples and do not limit the appended claims. The technology described in the appended claims includes various modifications and changes of the foregoing specific examples.

Claims

1. A secondary battery comprising: an electrode body including a positive electrode and a negative electrode; a battery case housing the electrode body; an electrode terminal attached to the battery case and in connection with the positive electrode or the negative electrode of the electrode body; and a gasket sandwiched between the battery case and the electrode terminal and sealing between the battery case and the electrode terminal, wherein the secondary battery is configured such that a parameter K calculated using the following equation (1):

$$K = W_i / (V_r G_{H_2}) \quad (1)$$
 (in the equation (1), W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm³) of void in the battery case, and G_{H_2} represents an H.sub.2 gas permeability coefficient (cm³·Math.cm)/(cm²·Math.s·Math.cmHg) of the gasket at 60° C.) satisfies 0.43×10^{-8} or more to 0.59×10^{-8} or less.

2. The secondary battery according to claim 1, wherein the H.sub.2 gas permeability coefficient G_{H_2} (cm³·Math.cm)/(cm²·Math.s·Math.cmHg) is 35.5×10^{-10} or more to 51.8×10^{-10} or less.

3. The secondary battery according to claim 1, further comprising a nonaqueous electrolyte including ethylene carbonate, ethyl methyl carbonate, and dimethyl carbonate.

4. A method of manufacturing a secondary battery comprising: an electrode body including a positive electrode and a negative electrode; a battery case housing the electrode body; an electrode terminal attached to the battery case; and a gasket sandwiched between the battery case and the electrode terminal and sealing between the battery case and the electrode terminal, the method comprising: setting a parameter K calculated using the following equation (1):

$$K = W_i / (V_r G_{H_2}) \quad (1)$$
 (in the equation (1), W_i represents an initial capacity (Ah) of the secondary battery, V_r represents a volume (cm³) of void in the battery case, and G_{H_2} represents an H.sub.2 gas permeability coefficient (cm³·Math.cm)/(cm²·Math.s·Math.cmHg) of the gasket at 60° C.) to satisfy 0.43×10^{-8} or more to 0.59×10^{-8} or less to construct the secondary battery.

5. The method according to claim 4, wherein the setting is performed with a gasket having the H.sub.2 gas permeability coefficient G_{H_2} (cm³·Math.cm)/(cm²·Math.s·Math.cmHg)

of 35.5×10^{-10} or more to 51.8×10^{-10} or less selected as the gasket.

6. The method according to claim 4, further comprising: injecting a nonaqueous electrolyte including ethylene carbonate, ethyl methyl carbonate, and dimethyl carbonate.
